

MISSOURI
S&T

MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 1.6

Applications of Linear Systems

Example 1

When solutions of sodium phosphate and barium nitrate are mixed, the result is barium phosphate (as a precipitate) and sodium nitrate. The unbalanced equation is

$$\text{Na}_3\text{PO}_4 + \text{Ba}(\text{NO}_3)_2 \rightarrow \text{Ba}_3(\text{PO}_4)_2 + \text{NaNO}_3$$

Balance the chemical equation.

Network Flow Problems

Basic idea: Flow in = Flow out

Example 2

Find the general traffic pattern of the freeway network shown in the figure. Flow rates are given in cars per minute.


